

The Structure of the Prime Lattice in the Classes $6k \pm 1$

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Abstract

All primes greater than three lie in the congruence classes $6k + 1$ and $6k - 1$. This document summarizes the structural properties of these two branches, showing how they form a closed multiplicative lattice. Composites generated from elements of the branches return to precise locations within the same branches, determined entirely by modular arithmetic. This feedback mechanism explains the appearance of composite intrusions such as 25, 35, 49, 55, 65, 77, 85, 91, 95, 115, 119, and 121 inside the prime lattice.

1 The Mod-6 Backbone

Every integer n satisfies $n \equiv 0, 1, 2, 3, 4, 5 \pmod{6}$. Only two of these residue classes can contain primes greater than 3:

$$n \equiv 1 \pmod{6}, \quad n \equiv 5 \pmod{6}.$$

All other classes are divisible by 2 or 3.

Thus the prime lattice is built on the backbone $6k$, $k \in \mathbb{Z}^+$, shifted by ± 1 .

2 The Two Prime Branches

We define the two branches:

$$B_1 = \{6k + 1 \mid k \geq 1\}, \quad B_5 = \{6k - 1 \mid k \geq 1\}.$$

These contain all primes greater than 3, and also contain certain composites generated by the multiplicative structure of the lattice.

3 Closure Under Multiplication

Let a and b be elements of the branches. Then

$$a \equiv \pm 1 \pmod{6}, \quad b \equiv \pm 1 \pmod{6}.$$

Their product satisfies

$$ab \equiv (\pm 1)(\pm 1) \equiv \pm 1 \pmod{6}.$$

Therefore every product of branch elements must fall back into one of the two branches. This is the closure property of the $6k \pm 1$ lattice.

4 Return Locations of Composites

A composite x generated from branch elements must satisfy $x = 6k \pm 1$. Its exact location is determined by solving either

$$6k + 1 = x \quad \text{or} \quad 6k - 1 = x.$$

This explains why composites appear at precise positions inside the branches.

4.1 Example: 25

We have

$$25 = 5 \times 5, \quad 5 \equiv -1 \pmod{6}.$$

Thus

$$(-1)(-1) = +1 \pmod{6},$$

so 25 must lie in B_1 . Solving

$$6k + 1 = 25 \implies k = 4,$$

shows that 25 is the fourth element of B_1 .

4.2 Example: 35

We have

$$35 = 5 \times 7, \quad 5 \equiv -1, \quad 7 \equiv +1 \pmod{6}.$$

Thus

$$(-1)(+1) = -1 \pmod{6},$$

so 35 must lie in B_5 . Solving

$$6k - 1 = 35 \implies k = 6,$$

shows that 35 is the sixth element of B_5 .

5 Composite Intrusions up to $m = 20$

The following composites appear in the first 20 elements of the branches:

$$B_1 : \{25, 49, 55, 85, 91, 115, 121\}, \quad B_5 : \{35, 65, 77, 95, 119\}.$$

Each arises from products of earlier branch elements and returns to its exact modular location.

6 Conclusion

The classes $6k \pm 1$ form a closed multiplicative lattice containing all primes greater than 3. Composites generated from branch elements return to precise positions determined by modular arithmetic. This structure explains the self-referential behavior of the prime lattice and provides a foundation for analyzing repeat phenomena such as pair, triple, and higher-order overlaps.

7 The Residue Flip and Backbone Reset Principle

The structure of the prime lattice in the classes $6k \pm 1$ is governed by a simple but fundamental rule: every element of the lattice is either congruent to $+1$ or -1 modulo 6. These two congruence classes form the branches

$$B_1 = 6k + 1, \quad B_5 = 6k - 1.$$

Theorem (Residue Flip and Backbone Reset)

Let a and b be elements of the branches B_1 or B_5 , so that

$$a \equiv \pm 1 \pmod{6}, \quad b \equiv \pm 1 \pmod{6}.$$

Then their product satisfies

$$ab \equiv (\pm 1)(\pm 1) \equiv \pm 1 \pmod{6}.$$

Thus every product of branch elements returns to the $6k \pm 1$ lattice. Moreover:

- If $ab \equiv +1 \pmod{6}$, then ab lies in B_1 .
- If $ab \equiv -1 \pmod{6}$, then ab lies in B_5 .

Interpretation

The multiplication of residues induces a “flip” between the two branches:

$$(+1)(+1) = +1, \quad (-1)(-1) = +1, \quad (+1)(-1) = -1.$$

Whenever the residue flips from $+1$ to -1 or vice versa, the resulting composite resets back to the $6k$ backbone and lands at the unique position determined by solving either

$$6k + 1 = ab \quad \text{or} \quad 6k - 1 = ab.$$

Consequences

This mechanism explains why composites such as 25, 35, 49, 55, 65, 77, 85, 91, 95, 115, 119, and 121 appear inside the branches. They are not arbitrary intrusions; they are the inevitable result of the residue flip and backbone reset within the closed multiplicative system generated by $6k \pm 1$.

8 Composite Return Location Lemma

Lemma (Composite Return Location)

Let x be a composite number formed by multiplying two elements of the prime branches B_1 and B_5 , where

$$B_1 = \{6k + 1 \mid k \geq 1\}, \quad B_5 = \{6k - 1 \mid k \geq 1\}.$$

Suppose

$$x = ab, \quad \text{with } a \equiv \pm 1 \pmod{6}, \quad b \equiv \pm 1 \pmod{6}.$$

Then x must satisfy

$$x \equiv \pm 1 \pmod{6},$$

and therefore x lies in exactly one of the two branches:

$$x = 6k + 1 \quad \text{or} \quad x = 6k - 1$$

for a unique integer k .

Proof

Since a and b are elements of B_1 or B_5 , their residues modulo 6 are either $+1$ or -1 . Thus

$$x = ab \equiv (\pm 1)(\pm 1) \equiv \pm 1 \pmod{6}.$$

No other residue class is possible. Therefore x must lie in one of the two congruence classes $6k + 1$ or $6k - 1$.

To determine the exact location of x in the lattice, we solve one of the equations

$$6k + 1 = x, \quad 6k - 1 = x.$$

Since x is fixed, each equation has at most one integer solution for k , and exactly one of them must hold because $x \equiv \pm 1 \pmod{6}$.

Thus every composite generated from branch elements returns to a unique position in the $6k \pm 1$ lattice.

Examples

- $25 = 5 \times 5$, with $5 \equiv -1 \pmod{6}$, so

$$(-1)(-1) = +1 \pmod{6}.$$

Thus 25 lies in B_1 and satisfies $25 = 6 \cdot 4 + 1$.

- $35 = 5 \times 7$, with $5 \equiv -1$ and $7 \equiv +1$, so

$$(-1)(+1) = -1 \pmod{6}.$$

Thus 35 lies in B_5 and satisfies $35 = 6 \cdot 6 - 1$.

Conclusion

Every composite formed from elements of B_1 and B_5 returns to a precise location in the $6k \pm 1$ lattice. This return mechanism is determined entirely by the residue flip law and the modular structure of the backbone $6k$.

9 The Backbone Reset Principle

Principle

The lattice formed by the congruence classes $6k \pm 1$ is supported by the integer backbone $6k$. Every element of the branches B_1 and B_5 is a shift of this backbone by $+1$ or -1 :

$$B_1 = 6k + 1, \quad B_5 = 6k - 1.$$

When two branch elements are multiplied, their residues modulo 6 determine the residue of the product:

$$(+1)(+1) = +1, \quad (-1)(-1) = +1, \quad (+1)(-1) = -1.$$

Thus the product ab always satisfies

$$ab \equiv \pm 1 \pmod{6}.$$

Backbone Reset

Since $ab \equiv \pm 1 \pmod{6}$, the composite ab must lie in one of the two branches. To determine its exact position, we “reset” the value ab to the backbone $6k$ by subtracting or adding 1:

$$ab = 6k + 1 \quad \text{or} \quad ab = 6k - 1.$$

Solving either equation yields a unique integer k , because the residue class of ab is already fixed. This reset mechanism ensures that every composite generated from branch elements returns to a precise location in the lattice.

Interpretation

The backbone reset principle states that the multiplicative structure of the prime lattice is closed: no composite formed from elements of B_1 or B_5 can escape the lattice. Instead, the residue flip determines whether the composite returns to B_1 or B_5 , and the backbone $6k$ determines the exact location.

Examples

- $25 = 5 \times 5$ satisfies $25 \equiv +1 \pmod{6}$, so it resets to $6k + 1$ and solves $25 = 6 \cdot 4 + 1$.
- $35 = 5 \times 7$ satisfies $35 \equiv -1 \pmod{6}$, so it resets to $6k - 1$ and solves $35 = 6 \cdot 6 - 1$.

Conclusion

The backbone reset principle explains why composites appear inside the branches at exact positions. The lattice $6k \pm 1$ is a closed system under multiplication, and the backbone $6k$ provides the unique coordinates for every composite generated within it.

10 The $12L \pm 1$ Composite Generation Lemma

Lemma (Squares and Cross-Branch Products at $12L \pm 1$)

Let n, a, b be positive integers and consider elements of the branches

$$B_1 = \{6k + 1 \mid k \geq 1\}, \quad B_5 = \{6k - 1 \mid k \geq 1\}.$$

Then:

- (1) For any branch element of the form $6n \pm 1$, its square satisfies

$$(6n - 1)^2 = 12L + 1, \quad (6n + 1)^2 = 12L + 1$$

for some integer L . In particular, every square of a branch element lies at a position of the form $12L + 1$ in B_1 .

- (2) For any cross-branch product of the form $(6a - 1)(6b + 1)$, we have

$$(6a - 1)(6b + 1) = 12L - 1$$

for some integer L . In particular, every such cross-branch composite lies at a position of the form $12L - 1$ in B_5 .

Proof

- (1) For squares, we compute

$$(6n - 1)^2 = 36n^2 - 12n + 1 = 12(3n^2 - n) + 1,$$

$$(6n + 1)^2 = 36n^2 + 12n + 1 = 12(3n^2 + n) + 1.$$

In both cases, the square is of the form $12L + 1$ with

$$L = 3n^2 - n \quad \text{or} \quad L = 3n^2 + n.$$

Thus every square of a branch element lies exactly at $12L + 1$, which is a $6k + 1$ position in B_1 .

- (2) For cross-branch products, we compute

$$(6a - 1)(6b + 1) = 36ab + 6a - 6b - 1.$$

Factor out 12:

$$36ab + 6a - 6b - 1 = 12 \left(3ab + \frac{a - b}{2} \right) - 1.$$

Since a and b are integers with $6a - 1$ and $6b + 1$ in the branches, the expression $3ab + \frac{a-b}{2}$ is an integer (this can be checked case by case for admissible a, b). Hence the product is of the form $12L - 1$ for some integer L , and therefore lies at a $6k - 1$ position in B_5 .

Examples

- $5^2 = 25 = 12 \cdot 2 + 1$ and $7^2 = 49 = 12 \cdot 4 + 1$ are squares of branch elements in B_5 and B_1 respectively, both landing at $12L + 1$.
- $5 \cdot 7 = 35 = 12 \cdot 3 - 1$ is a cross-branch product landing at $12L - 1$ in B_5 .

Conclusion

Squares of branch elements generate composites at positions $12L + 1$ in B_1 , while cross-branch products generate composites at positions $12L - 1$ in B_5 . These $12L \pm 1$ locations are the visible backbone feedback points where the multiplicative structure of the lattice re-enters the $6k \pm 1$ branches.

11 Per-column Feedback at $12L + 1$ for Branch Squares

Lemma (Guaranteed Feedback Locations for Squares)

Let c be a positive integer and consider branch elements

$$6c - 1 \in B_5, \quad 6c + 1 \in B_1.$$

Then:

- (1) The square of $6c - 1$ is

$$(6c - 1)^2 = 36c^2 - 12c + 1 = 12(3c^2 - c) + 1.$$

Thus, for

$$L_-(c) = 3c^2 - c,$$

the position

$$12L_-(c) + 1$$

is a guaranteed composite (feedback event), equal to $(6c - 1)^2$ and lying in B_1 .

- (2) The square of $6c + 1$ is

$$(6c + 1)^2 = 36c^2 + 12c + 1 = 12(3c^2 + c) + 1.$$

Thus, for

$$L_+(c) = 3c^2 + c,$$

the position

$$12L_+(c) + 1$$

is a guaranteed composite (feedback event), equal to $(6c + 1)^2$ and lying in B_1 .

Column Interpretation

If we index the backbone by k via

$$\text{backbone position} = 6k,$$

then at the feedback locations we have

$$k = L_-(c) \quad \text{or} \quad k = L_+(c),$$

so the column index k coincides with the L value generated by the branch parameter c . In other words, the feedback events occur at backbone columns

$$6k = 6L_-(c) \quad \text{or} \quad 6k = 6L_+(c),$$

and the corresponding branch entries

$$6k + 1 = (6c - 1)^2, \quad 6k + 1 = (6c + 1)^2$$

are composites produced by squaring the branch elements.

Examples

- For $c = 1$:

$$\begin{aligned} L_-(1) &= 3 \cdot 1^2 - 1 = 2, & 12L_-(1) + 1 &= 25 = 5^2, \\ L_+(1) &= 3 \cdot 1^2 + 1 = 4, & 12L_+(1) + 1 &= 49 = 7^2. \end{aligned}$$

- For $c = 2$:

$$\begin{aligned} L_-(2) &= 3 \cdot 4 - 2 = 10, & 12 \cdot 10 + 1 &= 121 = 11^2, \\ L_+(2) &= 3 \cdot 4 + 2 = 14, & 12 \cdot 14 + 1 &= 169 = 13^2. \end{aligned}$$

Conclusion

For each branch parameter c , there are two specific backbone columns $k = L_-(c)$ and $k = L_+(c)$ at which the lattice exhibits guaranteed feedback: the entries $6k + 1$ are composites equal to $(6c - 1)^2$ or $(6c + 1)^2$. These are the per-column backbone feedback points generated by squaring the branch elements $6c \pm 1$.

12 Cross-Branch Feedback at $12L - 1$

Lemma (Cross-Branch Composite Feedback)

Let c_1, c_2 be positive integers and consider branch elements

$$6c_1 - 1 \in B_5, \quad 6c_2 + 1 \in B_1.$$

Form the cross-branch product

$$x = (6c_1 - 1)(6c_2 + 1).$$

Then:

(1) The product expands to

$$(6c_1 - 1)(6c_2 + 1) = 36c_1c_2 + 6c_1 - 6c_2 - 1.$$

Factor out 12:

$$36c_1c_2 + 6c_1 - 6c_2 - 1 = 12 \left(3c_1c_2 + \frac{c_1 - c_2}{2} \right) - 1.$$

(2) If $c_1 - c_2$ is even, then

$$L(c_1, c_2) = 3c_1c_2 + \frac{c_1 - c_2}{2}$$

is an integer and

$$x = 12L(c_1, c_2) - 1.$$

Thus x lies at a $12L - 1$ position in B_5 and is a composite generated by cross-branch feedback.

Column Interpretation

If backbone columns are indexed by

$$\text{backbone position} = 6k,$$

then at these cross-branch feedback points we have

$$k = L(c_1, c_2),$$

and the corresponding branch entry

$$6k - 1 = 12L(c_1, c_2) - 1$$

is a composite equal to $(6c_1 - 1)(6c_2 + 1)$.

Examples

- $c_1 = 1, c_2 = 1$:

$$(6 \cdot 1 - 1)(6 \cdot 1 + 1) = 5 \cdot 7 = 35,$$

$$L(1, 1) = 3 \cdot 1 \cdot 1 + \frac{1 - 1}{2} = 3,$$

$$12L(1, 1) - 1 = 12 \cdot 3 - 1 = 35.$$

- $c_1 = 2, c_2 = 2$:

$$(11)(13) = 143,$$

$$L(2, 2) = 3 \cdot 2 \cdot 2 + \frac{2 - 2}{2} = 12,$$

$$12L(2, 2) - 1 = 12 \cdot 12 - 1 = 143.$$

Conclusion

Whenever $c_1 - c_2$ is even, the cross-branch product

$$(6c_1 - 1)(6c_2 + 1)$$

produces a composite at a backbone column $k = L(c_1, c_2)$, located at

$$6k - 1 = 12L(c_1, c_2) - 1 \in B_5.$$

These are the cross-branch backbone feedback points at $12L - 1$.

Python Implementation: Branch and Feedback Check

The following function tests primality by first ruling out composites caught by the square-feedback and cross-branch-feedback mechanisms above, then falling back to trial division over the remaining branch elements.

```
1 def is_prime_branch(n):
2     # Step 1: must be 6k+-1 except 2,3
3     if n in (2, 3):
4         return True
5     if n % 6 not in (1, 5):
6         return False
7
8     # Step 2: square feedback check
9     # Solve n = (6c+-1)^2
10    root = int(n**0.5)
11    if root*root == n:
12        if root % 6 in (1, 5):
13            return False # guaranteed composite square feedback
14
15    # Step 3: cross-branch feedback check
16    # Solve n = (6c1-1)(6c2+1)
17    # We only need to check c1,c2 up to sqrt(n)/6
18    limit = root // 6 + 2
19    for c1 in range(1, limit):
20        a = 6*c1 - 1
21        if a > root:
22            break
23        if n % a == 0:
24            b = n // a
25            if b % 6 == 1: # cross-branch partner
26                return False # guaranteed composite cross-branch
27                               feedback
28
29    # Step 4: fallback: check divisibility by earlier branch primes
30    # (this is the remaining sieve work)
31    for p in range(5, root + 1, 6):
```

```

31         if n % p == 0:
32             return False
33         if n % (p + 2) == 0:
34             return False
35
36     return True

```

13 Square Feedback at $12L + 1$

Lemma (Guaranteed Square Feedback)

Let c be a positive integer and consider the branch elements

$$6c - 1 \in B_5, \quad 6c + 1 \in B_1.$$

Then the squares of these branch elements always generate composites at positions of the form $12L + 1$.

(1) The square of $6c - 1$ is

$$(6c - 1)^2 = 36c^2 - 12c + 1 = 12(3c^2 - c) + 1.$$

Thus, for

$$L_-(c) = 3c^2 - c,$$

the position $12L_-(c) + 1$ is a guaranteed composite equal to $(6c - 1)^2$, lying in B_1 .

(2) The square of $6c + 1$ is

$$(6c + 1)^2 = 36c^2 + 12c + 1 = 12(3c^2 + c) + 1.$$

Thus, for

$$L_+(c) = 3c^2 + c,$$

the position $12L_+(c) + 1$ is a guaranteed composite equal to $(6c + 1)^2$, also lying in B_1 .

Column Interpretation

If backbone columns are indexed by $6k$, then at these feedback points we have

$$k = L_-(c) \quad \text{or} \quad k = L_+(c),$$

so the backbone column index k coincides with the L value generated by the branch parameter c . The corresponding branch entries

$$6k + 1 = (6c - 1)^2, \quad 6k + 1 = (6c + 1)^2$$

are guaranteed composite feedback events.

Examples

- For $c = 1$:

$$L_-(1) = 2, \quad 12 \cdot 2 + 1 = 25 = 5^2, \quad L_+(1) = 4, \quad 12 \cdot 4 + 1 = 49 = 7^2.$$

- For $c = 2$:

$$L_-(2) = 10, \quad 12 \cdot 10 + 1 = 121 = 11^2, \quad L_+(2) = 14, \quad 12 \cdot 14 + 1 = 169 = 13^2.$$

Conclusion

For each branch parameter c , the lattice exhibits two guaranteed feedback events at backbone columns $k = L_-(c)$ and $k = L_+(c)$, where the entries $6k + 1$ are composites equal to $(6c - 1)^2$ or $(6c + 1)^2$. These are the square-generated backbone feedback points at $12L + 1$.

14 Three-Band Branch Feedback Classification

Theorem (Branch–Band Feedback Structure)

Let

$$B_1 = \{6n + 1 \mid n \geq 1\}, \quad B_5 = \{6n - 1 \mid n \geq 1\}$$

denote the two multiplicative branches of the $6k \pm 1$ lattice. Every composite generated by branch elements $6c \pm 1$ lies in exactly one of the following three residue bands:

$$F_{+1} = \{12L + 1\}, \quad F_{+7} = \{12L + 7\}, \quad F_{-1} = \{12L - 1\}.$$

These bands correspond to three distinct feedback mechanisms in the lattice.

1. Square Feedback (Band F_{+1})

For any $c \geq 1$, the squares of branch elements satisfy

$$(6c - 1)^2 = 12(3c^2 - c) + 1 \in F_{+1},$$

$$(6c + 1)^2 = 12(3c^2 + c) + 1 \in F_{+1}.$$

Thus all square feedback events land in the $12L + 1$ band of B_1 .

2. Same-Branch Feedback (Bands F_{+1} and F_{+7})

Same-branch composites arise from

$$(6a + 1)(6b + 1), \quad (6a - 1)(6b - 1).$$

Expanding and reducing modulo 12 yields

$$(6a + 1)(6b + 1) \equiv 6(a + b) + 1 \pmod{12},$$

$$(6a - 1)(6b - 1) \equiv -6(a + b) + 1 \pmod{12}.$$

Hence:

$$(6a \pm 1)(6b \pm 1) \in \begin{cases} F_{+1}, & a + b \text{ even}, \\ F_{+7}, & a + b \text{ odd}. \end{cases}$$

All same-branch composites lie in B_1 , partitioned into the $12L + 1$ and $12L + 7$ bands according to the parity of $a + b$.

3. Cross-Branch Feedback (Band F_{-1})

Cross-branch composites arise from

$$(6a - 1)(6b + 1) = 36ab + 6a - 6b - 1.$$

Factoring 12 gives

$$(6a - 1)(6b + 1) = 12 \left(3ab + \frac{a - b}{2} \right) - 1.$$

Whenever $a - b$ is even, the expression in parentheses is an integer, and thus

$$(6a - 1)(6b + 1) \in F_{-1}.$$

These feedback events lie in the $12L - 1$ band of B_5 .

Conclusion

Every composite in the $6k \pm 1$ lattice arises from one of three feedback mechanisms:

- squares of branch elements (F_{+1}),
- same-branch products (F_{+1} or F_{+7}),
- cross-branch products (F_{-1}).

Thus the composite structure of the prime lattice is fully captured by the three residue bands $12L+1$, $12L+7$, and $12L-1$, each corresponding to a distinct multiplicative feedback pathway between the branches B_1 and B_5 .

Python Implementation: Full Lattice Feedback Check

This version checks all three feedback bands explicitly (square, same-branch, and cross-branch) via trial division over earlier branch elements.

```

1 def is_prime_lattice(n):
2     # Handle trivial small primes
3     if n in (2, 3):
4         return True
5
6     # Must be on the prime lattice 6k+-1

```

```

7  r = n % 6
8  if r not in (1, 5):
9      return False
10
11  # -----
12  # 1. Square feedback (12L+1 band)
13  # -----
14  root = int(n**0.5)
15  if root * root == n:
16      if root % 6 in (1, 5):
17          return False    # square of a branch element -> composite
18
19  # -----
20  # 2. Same-branch feedback (12L+1 or 12L+7 bands)
21  #   (6a+1)(6b+1) or (6a-1)(6b-1)
22  # -----
23  # Check divisibility by earlier branch elements
24  limit = root
25  for k in range(1, limit // 6 + 3):
26      a1 = 6*k + 1
27      a5 = 6*k - 1
28
29      # B1 x B1 same-branch composite
30      if a1 > 1 and a1 < n and n % a1 == 0:
31          return False
32
33      # B5 x B5 same-branch composite
34      if a5 > 1 and a5 < n and n % a5 == 0:
35          return False
36
37  # -----
38  # 3. Cross-branch feedback (12L-1 band)
39  #   (6a-1)(6b+1)
40  # -----
41  for k in range(1, limit // 6 + 3):
42      a = 6*k - 1
43      if a > 1 and a < n and n % a == 0:
44          b = n // a
45          if b % 6 == 1:    # cross-branch partner
46              return False
47
48  # -----
49  # If none of the feedback bands caught it -> PRIME
50  # -----
51  return True

```

Python Implementation: Band-Indexed Feedback Check

This version first computes the backbone index k and the band index L directly from n 's residues modulo 6 and 12, then tests only the feedback equations relevant to that band.

```
1 def is_prime_exclusive(n):
2     # Handle trivial primes
3     if n in (2, 3):
4         return True
5
6     # Must be  $6k+1$ 
7     r6 = n % 6
8     if r6 not in (1, 5):
9         return False
10
11     # Compute backbone index k
12     if r6 == 1:
13         k = (n - 1) // 6
14     else:
15         k = (n + 1) // 6
16
17     # Compute band index L
18     r12 = n % 12
19     if r12 == 1:
20         L = (n - 1) // 12
21     elif r12 == 7:
22         L = (n - 7) // 12
23     elif r12 == 11:
24         L = (n + 1) // 12
25     else:
26         # Not in any feedback band -> PRIME
27         return True
28
29     # -----
30     # 1. Square feedback:  $n = (6c+1)^2$ 
31     # -----
32     c = int((n**0.5 + 1) // 6)
33     if (6*c - 1)**2 == n or (6*c + 1)**2 == n:
34         return False
35
36     # -----
37     # 2. Same-branch feedback
38     # Check if  $n = (6a+1)(6b+1)$  or  $(6a-1)(6b-1)$ 
39     # -----
40     # Solve for possible a,b from  $n = 36ab \pm 6(a+b) + 1$ 
41     # We only need to check a up to  $\sqrt{n}/6$ 
42     limit = int((n**0.5) // 6) + 2
43
```

```

44     for a in range(1, limit):
45         A1 = 6*a + 1
46         A5 = 6*a - 1
47
48         if n % A1 == 0:
49             b = n // A1
50             if b % 6 == 1:  # B1 x B1
51                 return False
52
53         if n % A5 == 0:
54             b = n // A5
55             if b % 6 == 5:  # B5 x B5
56                 return False
57
58     # -----
59     # 3. Cross-branch feedback: n = (6a-1)(6b+1)
60     # -----
61     for a in range(1, limit):
62         A = 6*a - 1
63         if n % A == 0:
64             b = n // A
65             if b % 6 == 1:  # cross-branch partner
66                 return False
67
68     # -----
69     # If none of the feedback equations match -> PRIME
70     # -----
71     return True

```

15 Alignment Theorem for the Prime Lattice Branches

Let

$$B_1 = \{6k + 1 \mid k \geq 1\}, \quad B_5 = \{6k - 1 \mid k \geq 1\}$$

denote the two multiplicative branches of the prime lattice. Define the shifted branch

$$B_5^{(-s)}(k) = B_5(k + s),$$

that is, the branch B_5 shifted left by s positions.

Theorem 1 (Branch Alignment Theorem). *For the lattice branches B_1 and B_5 , the following structural alignments hold:*

1. *Same-branch composite corridor (shift -1). Shifting B_5 by one position,*

$$B_5^{(-1)}(k) = 6(k + 1) - 1,$$

aligns the values

$$(6k + 1, 6(k + 1) - 1)$$

so that all same-branch composite intrusions

$$25, 35, 49, 55, 85, 91, 115, 121, \dots$$

appear on the diagonal corridor formed by the pairs

$$(B_1(k), B_5^{(-1)}(k)).$$

2. Mod-5 alignment (shift -2). Shifting B_5 by two positions,

$$B_5^{(-2)}(k) = 6(k + 2) - 1,$$

aligns all values congruent to 5 (mod 12) in B_5 with the corresponding positions in B_1 . That is,

$$B_5(k) \equiv 5 \pmod{12} \iff B_5^{(-2)}(k) \text{ aligns with } B_1(k).$$

This produces the structural alignment

$$(7, 11), (13, 17), (19, 23), (25, 29), (31, 35), \dots$$

in which all mod-5 positions fall on a single diagonal.

3. Mod-7 alignment (shift -5). Shifting B_5 by five positions,

$$B_5^{(-5)}(k) = 6(k + 5) - 1,$$

aligns all values congruent to 7 (mod 12) in B_1 with the corresponding values in B_5 . Explicitly,

$$B_1(k) \equiv 7 \pmod{12} \iff B_5^{(-5)}(k) \text{ aligns with } B_1(k).$$

This produces the diagonal

$$(7, 23), (13, 29), (19, 35), (25, 41), (31, 47), \dots$$

containing all mod-7 positions.

These three alignments reveal the internal geometric structure of the prime lattice: each residue band $12L + 1$, $12L + 5$, and $12L + 7$ forms a distinct diagonal corridor obtained by a fixed shift of the B_5 branch. Composite intrusions occur precisely on these corridors, reflecting the closed multiplicative feedback of the lattice.

16 Composite Detector Theorem in the Prime Lattice

Let

$$B_1 = \{6k + 1 \mid k \geq 1\}, \quad B_5 = \{6k - 1 \mid k \geq 1\}$$

denote the two branches of the $6k \pm 1$ prime lattice. Define the index set

$$K = \{1, 2, \dots, m\}$$

for some fixed cutoff $m \in \mathbb{N}$, and the corresponding branch vectors

$$B_1(K) = \{6k + 1 \mid k \in K\}, \quad B_5(K) = \{6k - 1 \mid k \in K\}.$$

Theorem 2 (Vectorized Composite Detector). *All composites in the truncated lattice $B_1(K) \cup B_5(K)$ are generated by the following three feedback mechanisms:*

1. **Square feedback.** For each $k \in K$, define

$$s_1(k) = (6k + 1)^2, \quad s_5(k) = (6k - 1)^2.$$

The square feedback set is

$$S = \{s_1(k), s_5(k) \mid k \in K\}.$$

2. **Same-branch feedback.** For each pair $(a, b) \in K \times K$, define

$$p_{11}(a, b) = (6a + 1)(6b + 1), \quad p_{55}(a, b) = (6a - 1)(6b - 1).$$

The same-branch feedback set is

$$P_{sb} = \{p_{11}(a, b), p_{55}(a, b) \mid a, b \in K\}.$$

3. **Cross-branch feedback.** For each pair $(a, b) \in K \times K$, define

$$p_{51}(a, b) = (6a - 1)(6b + 1).$$

The cross-branch feedback set is

$$P_{cb} = \{p_{51}(a, b) \mid a, b \in K\}.$$

Let

$$C = S \cup P_{sb} \cup P_{cb}$$

denote the global feedback composite set. Then the composites in the truncated branches are given by the vectorized membership conditions

$$\text{Comp}(B_1(K)) = B_1(K) \cap C, \quad \text{Comp}(B_5(K)) = B_5(K) \cap C.$$

Equivalently, an element $x \in B_1(K) \cup B_5(K)$ is composite if and only if

$$x \in S \cup P_{sb} \cup P_{cb},$$

and it is prime if and only if

$$x \in B_1(K) \cup B_5(K) \quad \text{and} \quad x \notin S \cup P_{sb} \cup P_{cb}.$$

Python Implementation: Vectorized Composite Detector

This implementation computes all three feedback sets directly as NumPy arrays, avoiding any per-element loop.

```
1 import numpy as np
2
3 def composite_detector(kmax):
4     # Branch vectors
5     k = np.arange(1, kmax+1)
6     B1 = 6*k + 1
7     B5 = 6*k - 1
8
9     # --- 1. Square feedback -----
10    sq1 = (6*k + 1)**2
11    sq5 = (6*k - 1)**2
12    squares = np.concatenate([sq1, sq5])
13
14    # --- 2. Same-branch feedback -----
15    # B1 x B1
16    A1 = 6*k + 1
17    SB11 = np.outer(A1, A1).flatten()
18
19    # B5 x B5
20    A5 = 6*k - 1
21    SB55 = np.outer(A5, A5).flatten()
22
23    same_branch = np.concatenate([SB11, SB55])
24
25    # --- 3. Cross-branch feedback -----
26    cross = np.outer(A5, A1).flatten()
27
28    # --- Combine all feedback composites -----
29    composites = np.unique(
30        np.concatenate([squares, same_branch, cross])
31    )
32
33    # --- Detect composites in B1 and B5 -----
34    C1 = B1[np.isin(B1, composites)]
35    C5 = B5[np.isin(B5, composites)]
36
37    return B1, B5, C1, C5
38
39 # Example
40 B1, B5, C1, C5 = composite_detector(30)
41 print("Composites in B1:", C1)
42 print("Composites in B5:", C5)
```

17 Companion Mapping Theorem in the $6k \pm 1$ Lattice

Theorem 3 (Companion Mapping in the $6k \pm 1$ Lattice). *Let a be an integer of the form*

$$a = 6m \pm 1, \quad m \in \mathbb{Z},$$

and let

$$b_k = 6k \pm 1, \quad k \in \mathbb{Z}_{\geq 1}.$$

Define the companion set of a by

$$C(a) = \{ab_k : b_k = 6k \pm 1, \ k \geq 1\}.$$

Then:

1. **Closure in the lattice.** *Every companion $c \in C(a)$ is again of the form*

$$c = 6n \pm 1$$

for some integer n . In particular, $C(a)$ is contained in the $6k \pm 1$ lattice.

2. **Composite generation.** *If $|a| > 1$, then every $c \in C(a)$ is composite and has a as a non-trivial factor.*

3. **Factor family.** *For fixed a , the set $C(a)$ is exactly the set of all lattice elements that are divisible by a and have a cofactor in the lattice:*

$$C(a) = \{n \in \mathbb{Z} : n = ab, \ b = 6k \pm 1, \ k \geq 1\}.$$

Proof

Let $a = 6m + \varepsilon$ and $b_k = 6k + \delta$, where $\varepsilon, \delta \in \{-1, 1\}$. Then

$$c = ab_k = (6m + \varepsilon)(6k + \delta) = 36mk + 6m\delta + 6k\varepsilon + \varepsilon\delta.$$

Factor out 6:

$$c = 6(6mk + m\delta + k\varepsilon) + \varepsilon\delta.$$

Since $\varepsilon, \delta \in \{-1, 1\}$, we have $\varepsilon\delta \in \{-1, 1\}$. Thus

$$c = 6n \pm 1, \quad n = 6mk + m\delta + k\varepsilon \in \mathbb{Z},$$

which proves closure in the $6k \pm 1$ lattice.

If $|a| > 1$, then $c = ab_k$ has a non-trivial factor a , so c is composite, proving composite generation.

Finally, any lattice element n divisible by a with cofactor b in the lattice must satisfy

$$n = ab, \quad a = 6m \pm 1, \quad b = 6k \pm 1,$$

so $n \in C(a)$. Conversely, every element of $C(a)$ is of this form, proving the factor family characterization. $\square$

Python Implementation: Companion Set

```
1 k_values = np.arange(1, 100+1)
2 def companions(a, k_values):
3     # a is a lattice element, k_values is a list of k
4     return [a * (6*k - 1) for k in k_values] + [a * (6*k + 1) for k
5         in k_values]
6 print(companions(25, k_values))
```